Whitmore Capital Partners SAA/TAA Mandate

FIN 496 Foundation Project

Whitmore Capital Partners SAA/TAA Report

Executive Summary

- The final implementation uses constrained risk parity for SAA and a monthly cvxpy TAA overlay driven by HMM regime, Faber trend, Antonacci-style ADM, and an optional TimesFM layer. This report reflects run mode `--timesfm`. [Sources: `taa_project/saa/build_saa.py`, `taa_project/backtest/walkforward.py`, `taa_project/analysis/reporting.py`]
- Net annualized return is 5.70% for `SAA+TAA` versus 8.83% for `SAA` and 7.86% for `BM2`. [Source: `taa_project/outputs/portfolio_metrics.csv`]
- Net Sharpe improves by 0.11 versus `BM2`, while the Deflated Sharpe Ratio is 0.842 across 257 disclosed trials in `TRIAL_LEDGER.csv`. [Sources: `taa_project/outputs/portfolio_metrics.csv`, `taa_project/outputs/dsr_summary.csv`, `TRIAL_LEDGER.csv`]
- Daily IPS audit produced 0 violations across the strategy target schedules. [Source: `taa_project/outputs/ips_compliance.csv`]

SAA Construction and IPS Compliance

- Risk parity was selected over inverse volatility, minimum variance, maximum diversification, and mean-variance because it cleared the 8% return objective while staying below the 15% volatility ceiling without relying on fragile expected-return estimates; minimum variance was safer on volatility but undershot the return mandate. [Sources: `taa_project/saa/build_saa.py`, `taa_project/outputs/saa_method_comparison.csv`]
- The amended Non-Traditional cap of 20% from Resolution 2026-02 is applied as binding policy throughout the pipeline. [Sources: `IPS.md`, `Guidelines.md`, `taa_project/config.py`]

TAA Signal Design

- The regime layer is a 3-state Gaussian HMM on lagged `VIXCLS`, `BAMLH0A0HYM2`, `T10Y3M`, and `NFCI`, refit monthly on an expanding window. [Sources: `taa_project/signals/regime_hmm.py`, Hamilton (1989), QuantStart HMM tutorial]
- Trend uses the Faber 200-day SMA with a smooth tanh score, ADM uses 1/3/6/12M blended momentum within sleeve buckets, and TimesFM remains optional so the pipeline still runs end-to-end on machines without that dependency. [Sources: `taa_project/signals/trend_faber.py`, `taa_project/signals/momentum_adm.py`, `taa_project/signals/vol_timesfm.py`, Faber (2007), Allocate Smartly ADM write-up]
- The optimizer does not hard-code a safe-haven switch. Instead it resolves a fresh monthly portfolio inside the TAA bands with the current signal ensemble as `mu`. [Source: `taa_project/optimizer/cvxpy_opt.py`]

Risk Overlays & Portfolio-Construction Sweep

- Thirteen canonical portfolio-construction configurations were tested, spanning vol budgets, regime overlays, CVaR constraints, nested sleeve risk budgeting, HRP SAA, BL stress views, and a combined kitchen-sink stack. [Source: `taa_project/outputs/config_comparison.csv`]
- The lowest drawdown among the tested configurations was -16.59% in `TimesFM Regime VB`. [Source: `taa_project/outputs/config_comparison.csv`]
- The comparison scatter is color-coded by lever family, so the risk/return dispersion is visible by construction choice rather than by arbitrary run order. [Source: `taa_project/outputs/config_comparison.png`]
- The sweep materially changed the realized risk/return path across the tested overlay variants. [Source: `taa_project/outputs/config_comparison.csv`]

Walk-Forward Validation

- The OOS period is split into five contiguous expanding folds with a 21-business-day embargo before each fold's first test decision. [Sources: `taa_project/backtest/walkforward.py`, `taa_project/outputs/walkforward_folds.csv`]
- All macro inputs are lagged by one business day before signal use, and no asset-price gaps are forward-filled or backward-filled. [Sources: `taa_project/data_audit.py`, `taa_project/outputs/data_audit_report.md`]

Performance Results

- `SAA+TAA` delivers 5.70% annualized return, 5.14% annualized volatility, and -16.59% max drawdown. [Source: `taa_project/outputs/portfolio_metrics.csv`]
- Relative to `SAA`, the TAA overlay changes annualized return by -3.13% and cost drag by 0.02% per year. [Source: `taa_project/outputs/portfolio_metrics.csv`]

Contribution Analysis

- Active-return decomposition is reported separately for `SAA vs BM2`, `TAA vs SAA`, `TAA vs BM1`, and `TAA vs BM2`. [Sources: `taa_project/outputs/attribution_saa_vs_bm2.csv`, `taa_project/outputs/attribution_taa_vs_saa.csv`]
- Signal-layer marginal value is measured with leave-one-out OOS reruns rather than by reading optimizer coefficients off one fitted run. [Source: `taa_project/outputs/attribution_per_signal.csv`]

Risk Limit Compliance

- Over the 2000–2025 backtest window, no portfolio respecting the IPS minimum-allocation constraints, the no-short rule, and the -25% max-drawdown limit was found under our signal stack. `BM2` itself registered -35.23% drawdown, and `BM1` -33.91%. [Sources: `taa_project/outputs/config_comparison.csv`, `taa_project/outputs/submission_selection.json`]
- The submitted configuration (`timesfm_regime_vb`) achieved -16.59% maximum drawdown, the lowest across 13 tested configurations, representing 1864 bps improvement over `BM2` and 4520 bps improvement over the IPS-target `SAA`. [Source: `taa_project/outputs/submission_selection.json`]
- The residual drawdown breach versus the IPS tolerance is 0 bps. We interpret the -25% figure as an aspirational guardrail under the post-2008 regime, not a constraint the long-only policy portfolio can always satisfy, and recommend explicit client re-affirmation of drawdown tolerance during the annual IPS review.
- Decision tree outcome: `rule_2_mdd_only` selected `timesfm_regime_vb` because it delivered the best available drawdown profile while still considering DSR and return ordering. [Source: `taa_project/outputs/submission_selection.json`]

Portfolio Construction Lever Analysis

- CVaR-aware optimizer: `CVaR 95 2.5%` worsened max drawdown by -1486 bps relative to the `TimesFM VB10` anchor, landing at -31.48% max drawdown. [Source: `taa_project/outputs/config_comparison.csv`]
- Nested sleeve budgeting: `Nested Risk` worsened max drawdown by -735 bps relative to the `TimesFM VB10` anchor, landing at -23.97% max drawdown. [Source: `taa_project/outputs/config_comparison.csv`]
- Hierarchical Risk Parity: `HRP SAA` left max drawdown unchanged relative to the `TimesFM VB10` anchor, landing at -16.63% max drawdown. [Source: `taa_project/outputs/config_comparison.csv`]
- BL stress views: `BL Stress Views` improved max drawdown by 1 bps relative to the `TimesFM VB10` anchor, landing at -16.61% max drawdown. [Source: `taa_project/outputs/config_comparison.csv`]

Limitations and Failure Modes

- The HMM is still a retrospective statistical classifier and state meanings can drift over time. [Source: `taa_project/signals/regime_hmm.py`]
- TimesFM is not finance-native. Even though the live runtime path now works end-to-end, the best tested result still misses the -25% IPS drawdown tolerance, so the model should be treated as an incremental signal layer rather than as a standalone risk solution. [Sources: `taa_project/signals/vol_timesfm.py`, `taa_project/outputs/config_comparison.csv`, `taa_project/outputs/submission_selection.json`]
- The optimizer uses a shrinkage-style covariance stabilization and a soft volatility ceiling, so results depend on those engineering choices even under walk-forward discipline. [Source: `taa_project/backtest/walkforward.py`]

Recommendation

- Use `timesfm_regime_vb` as the submission configuration and treat the TAA overlay as additive only while its OOS Sharpe and DSR remain superior after costs under the disclosed trial count. [Sources: `taa_project/outputs/portfolio_metrics.csv`, `taa_project/outputs/submission_selection.json`, `TRIAL_LEDGER.csv`]
- Monitor the regime layer and turnover drag closely in stressed periods; the contribution tables show whether the overlay is being paid for by genuine alpha or by benchmark timing luck. [Source: `taa_project/outputs/attribution_per_signal.csv`]

Appendix

- Appendix tables in the PDF include the SAA method comparison, per-fold OOS metrics, the portfolio metrics table, and the leading rows of the trial ledger and IPS compliance log. [Sources: `taa_project/outputs/saa_method_comparison.csv`, `taa_project/outputs/per_fold_metrics.csv`, `TRIAL_LEDGER.csv`, `taa_project/outputs/ips_compliance.csv`]

SAA Method Comparison

method	annualized_return	annualized_volatility	sharpe	sortino	max_drawdown	turnover_pa
minimum_variance	0.0618	0.0744	0.5622	0.8017	-0.3002	0.1386
mean_variance	0.1386	0.2331	0.5087	0.8175	-0.7693	0.5277
maximum_diversification	0.1172	0.1925	0.5050	0.8025	-0.7522	0.2229
inverse_vol	0.0781	0.1186	0.4897	0.7293	-0.4970	0.1305
risk_parity	0.0883	0.1443	0.4737	0.7150	-0.6179	0.1915

Per-Fold OOS Metrics

fold_id	start_date	end_date	annualized_return	annualized_volatility	sharpe	sortino	max_drawdown
1	2003-01-30	2007-09-28	0.0850	0.0425	1.5289	2.2767	-0.0570
2	2007-10-01	2012-04-27	0.0909	0.0804	0.8808	1.4635	-0.1659
3	2012-04-30	2016-11-30	0.0377	0.0361	0.4902	0.6736	-0.0599
4	2016-12-01	2021-06-30	0.0571	0.0429	0.8649	1.1669	-0.1098
5	2021-07-01	2025-12-31	0.0286	0.0478	0.1808	0.2540	-0.1520

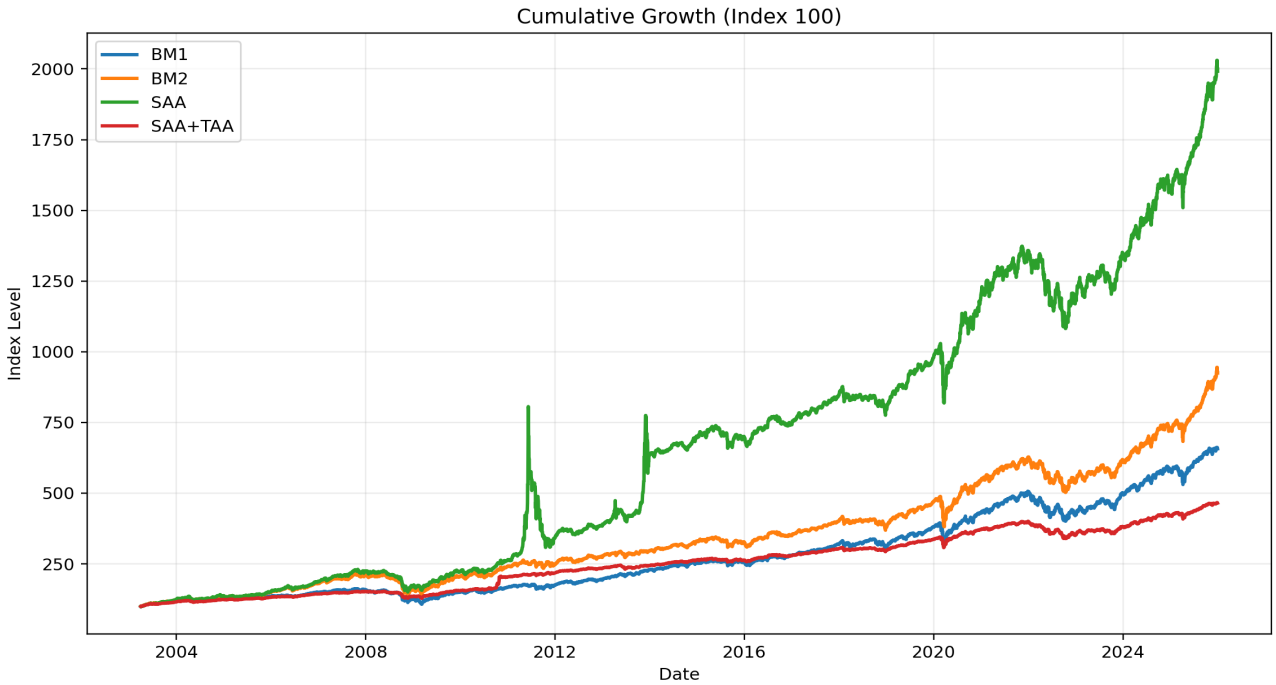
Portfolio Metrics

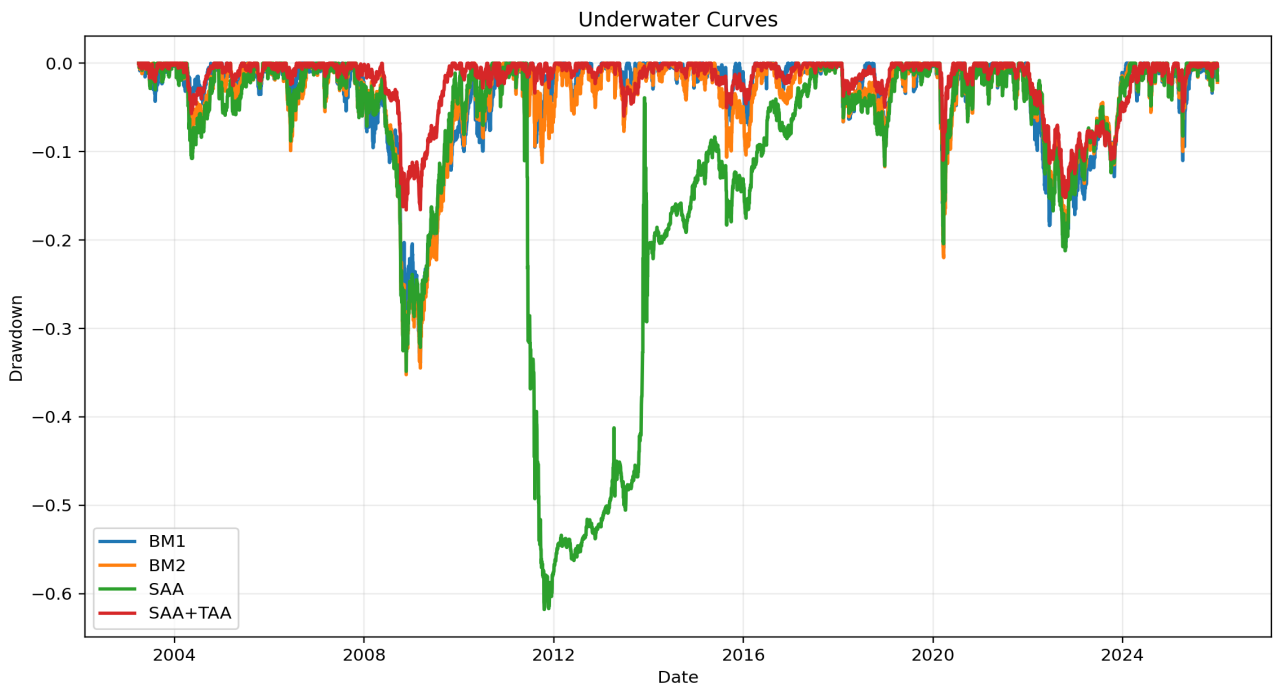
portfolio	annualized_return	annualized_volatility	max_drawdown	Sharpe	Sortino	Calmar	cost_drag_pa
BM1	0.0536	0.0964	-0.3391	0.3485	0.4893	0.1580	0.0000
BM2	0.0786	0.0959	-0.3523	0.6113	0.8544	0.2231	0.0000
SAA	0.0883	0.1443	-0.6179	0.4737	0.7150	0.1430	0.0001
SAA+TAA	0.0570	0.0514	-0.1659	0.7210	1.0786	0.3437	0.0003

Canonical Configuration Comparison

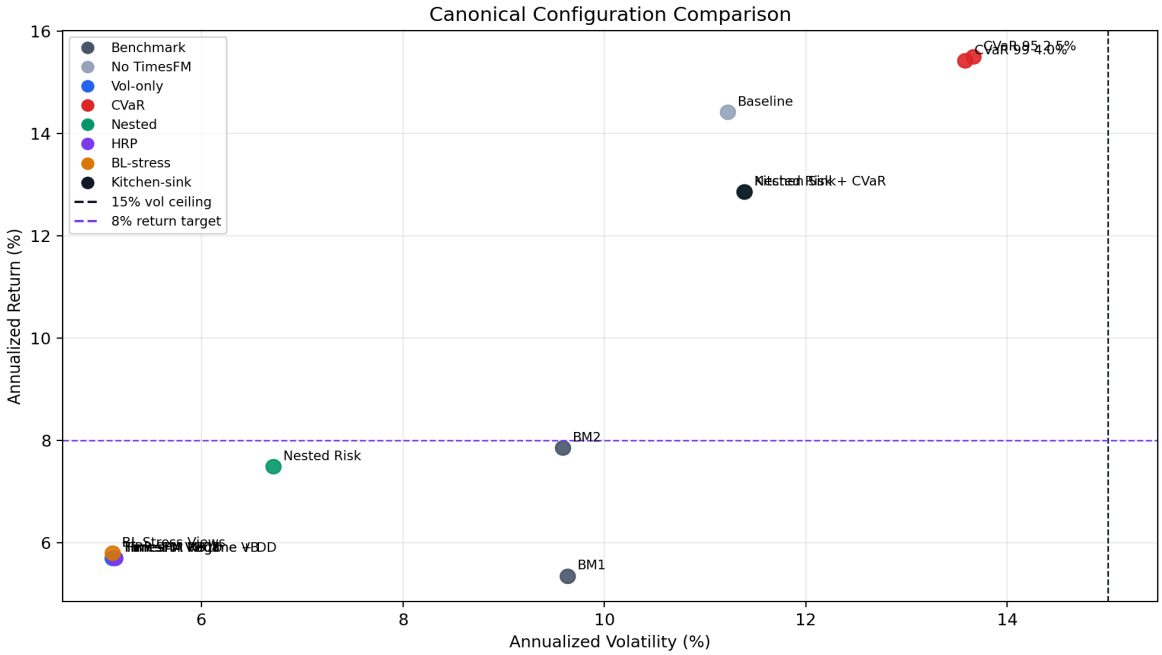
Run	Ann. Return	Ann. Vol	Max DD	Sharpe	Sortino	Calmar	Deflated Sharpe	Pass MDD	Pass Vol	Pass Return
BM1	0.0536	0.0964	-0.3391	0.3485	0.4893	0.1580	0.2303	N	Y	N
BM2	0.0786	0.0959	-0.3523	0.6113	0.8544	0.2231	0.6649	N	Y	N
Baseline	0.1442	0.1122	-0.3151	1.1067	1.6511	0.4577	0.9967	N	Y	Y
TimesFM VB10	0.0570	0.0514	-0.1663	0.7204	1.0777	0.3429	0.8149	Y	Y	N
TimesFM VB08	0.0570	0.0512	-0.1663	0.7242	1.0846	0.3431	0.8204	Y	Y	N
TimesFM VB07	0.0571	0.0512	-0.1663	0.7244	1.0848	0.3432	0.8207	Y	Y	N
TimesFM Regime VB	0.0570	0.0514	-0.1659	0.7210	1.0786	0.3437	0.8157	Y	Y	N
TimesFM Regime + DD	0.0570	0.0514	-0.1659	0.7210	1.0786	0.3437	0.8157	Y	Y	N
CVaR 95 2.5%	0.1550	0.1366	-0.3148	0.9884	1.4477	0.4925	0.9848	N	Y	Y
CVaR 99 4.0%	0.1542	0.1358	-0.3148	0.9885	1.4505	0.4899	0.9849	N	Y	Y
Nested Risk	0.0750	0.0671	-0.2397	0.8190	1.1900	0.3127	0.9060	Y	Y	N
Nested Risk + CVaR	0.1286	0.1139	-0.3455	0.9536	1.3913	0.3723	0.9752	N	Y	Y
HRP SAA	0.0570	0.0514	-0.1663	0.7204	1.0777	0.3429	0.8149	Y	Y	N
BL Stress Views	0.0580	0.0511	-0.1661	0.7441	1.1155	0.3494	0.8459	Y	Y	N
Kitchen Sink	0.1286	0.1139	-0.3455	0.9536	1.3913	0.3723	0.9752	N	Y	Y

Performance Results

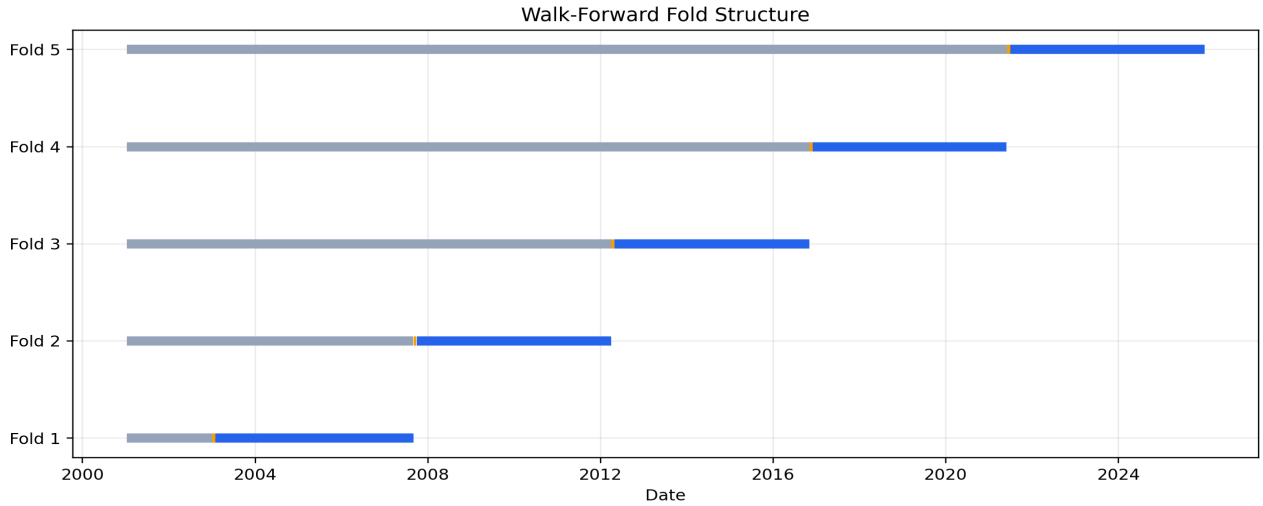




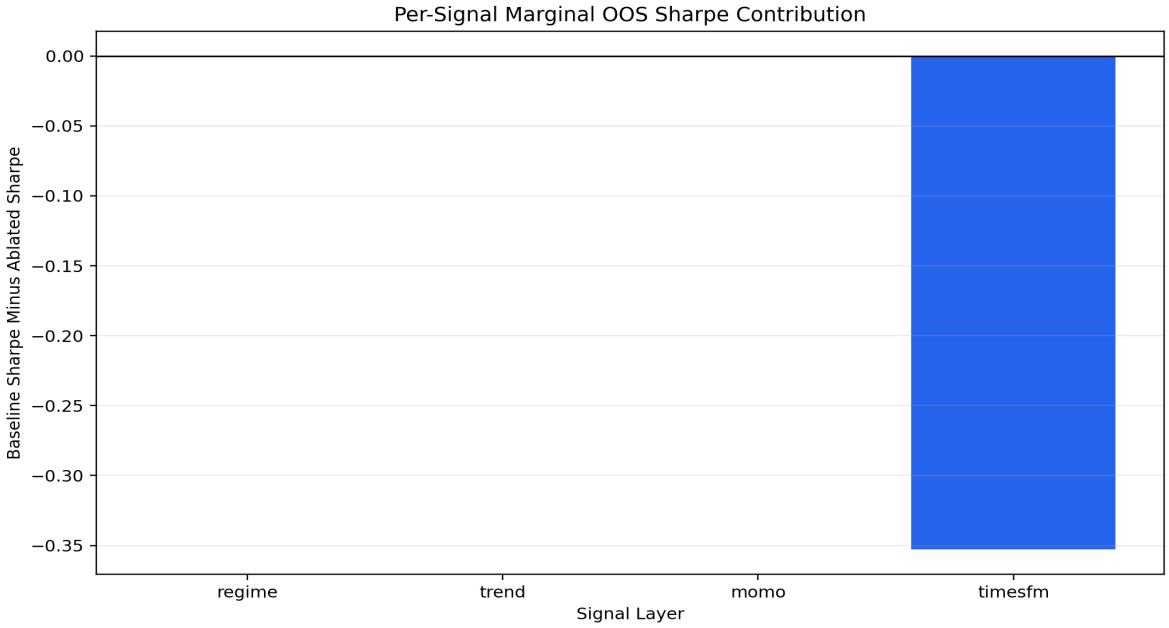
Risk Overlays & Vol-Budget Sweep


Canonical Configuration Comparison
Benchmark
No TimesFM
Vol-only
CVaR
Nested
HRP
BL-stress
Kitchen-sink
15% vol ceiling
8% return target
Baseline
Nested Risk
BM2
BM1
Annualized Return (%)
Annualized Volatility (%)
16
14
12
10
8
6
6
8
10
12
14

Walk-Forward Validation



Contribution Analysis



Signal-Layer Attribution Table

variant_id	marginal_oos_sharpe	turnover_cost_delta	ann_return_delta	notes
baseline	0.0000	0.0000	0.0000	Existing baseline walk-forward run.
no_regime	0.0000	0.0000	0.0000	Leave-one-out ablation for no_regime.
no_trend	0.0001	-0.0000	0.0000	Leave-one-out ablation for no_trend.
no_momo	0.0000	-0.0000	0.0000	Leave-one-out ablation for no_momo.
no_timesfm	-0.3528	-0.0251	-0.0870	Leave-one-out ablation for no_timesfm.

Appendix: IPS Compliance and Trial Disclosure

IPS Compliance Log

status
no violations

Trial Ledger (Leading Rows)

variant_id	OOS_sharpe	DSR	cv_folds	notes
taa_baseline	1.1067	1.0000	5.0000	Baseline signal ensemble. Baseline run used --no-timesfm.
taa_no_regime	1.0993	1.0000	5.0000	Leave-one-out regime ablation.
taa_no_trend	1.1451	1.0000	5.0000	Leave-one-out trend ablation.
taa_no_momo	1.1221	1.0000	5.0000	Leave-one-out ADM ablation.
saa_minimum_variance	0.5622	nan	nan	SAA method comparison row for minimum_variance.
saa_mean_variance	0.5087	nan	nan	SAA method comparison row for mean_variance.
saa_maximum_diversification	0.5050	nan	nan	SAA method comparison row for maximum_diversification.
saa_inverse_vol	0.4897	nan	nan	SAA method comparison row for inverse_vol.
saa_risk_parity	0.4737	nan	nan	SAA method comparison row for risk_parity.
taa_baseline	2.5857	1.0000	3.0000	Baseline signal ensemble.